

# Integrating Family Privacy Education and Informal Learning Spaces: Characteristics, Challenges and Design Opportunities

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## ABSTRACT

Children face increasing privacy risks and the need to navigate complex choices, while privacy education is not sufficient due to limited education scope and family involvement. We advocate for informal learning spaces (ILS) as a pioneering channel for family-based privacy education, given their established role in holistic technology and digital literacy education, which specifically targets family groups. In this paper, we conducted an interview study with five families to understand revealing current approaches to privacy education and engagement with ILS for family-based learning. Our findings highlight ILS's trans-formative potential in family privacy education, considering existing practices and challenges. We discuss the reason for family-based privacy education in ILS and identify potential design opportunities. Additionally, we outline our future work, which involves expanding participant involvement and conducting co-design activities with family groups to create design prototypes.

## CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability.

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## 1 INTRODUCTION

Children (6-14 years old) are getting more attached to various technologies, from social media to handheld electronic devices, to wearable devices and other emerging technologies. Yet, their privacy knowledge and their ability to make informed privacy decisions do not match with technological advancement. Research has suggested that while children have a basic understanding of fundamental terminologies, they lack essential knowledge and ability to foster decision-making skills and take responsible digital behaviors [17, 34]. For example, children are often asked to follow certain rules [14], yet the rule-based approach oftentimes does not

suffice the varying privacy needs given the contextual nature of privacy [26].

As such, researchers have been exploring different means to carry out privacy education for children to improve their privacy knowledge and foster their skills to make informed privacy decisions. For example, an interactive medium is designed to foster early adolescents' awareness of information privacy in their everyday online practices [38]. There are also some games designed to develop children's and teenagers' privacy literacy [21, 29]

However, we believe that these existing practices of privacy education are limited mainly in the following three ways. First, most privacy education practices predominantly focused on online privacy education [7, 35, 41] and thus were not effective in addressing privacy risks in the offline space or in the intersection of online and offline contexts. This is particularly important considering children's increasing engagement with various cyber-physical devices that span across both real-world and online spaces (e.g., smart home devices, virtual reality, augmented reality, etc.), which were known for posing significant privacy risks to their users [1, 15, 37].

Secondly, current privacy education approaches focus on addressing privacy issues through one-time, short educational activities (e.g., using mobile applications [44] or interactive e-books [42]) rather than improving children's privacy literacy. These methods offer instant learning content and feedback, lacking continual reminders and practice in children's daily lives. As a result, these approaches fall short of generating a long-term impact on children's privacy knowledge and behavior. Thirdly, existing privacy education approaches primarily target individuals through school learning or self-learning [5, 13, 28, 40] and do not leverage their immediate family members in this process [23]. Such omission could negatively impact their family members' ability to guide their children in making privacy decisions [20].

In searching for ways to bridge these gaps, we noticed that the HCI community has extensively explored utilizing informal learning spaces (ILS), such as museums, science centers, libraries, for technology and digital literacy education [18], many of which target family groups, a prevalent demographic in these settings [6, 11, 25]. This line of research inspired us to think, **is it possible to utilize ILS for family-based privacy education to improve children's privacy literacy?**

As privacy researchers and educators, our goal is to explore a novel, **family-based privacy education approach in ILS**. As family often experiences enriched interactions in informal learning settings [8, 24, 30], we aim to build interactive, privacy-centric exhibits or artifacts and display them in various ILS. Ideally, interacting with these exhibits or artifacts would spark family-based discussions around privacy and expose several "teachable moments"

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for us to inject relevant privacy education materials for deeper learning.

However, this novel approach at the intersection of family-based privacy education and informal learning has not been studied in the literature. On the one hand, it is not clear how families go about privacy education especially considering the possible imbalanced power dynamics and conflicted interests, hindering the collaborative practices of addressing privacy concerns [2, 12, 16, 43]. On the other hand, it is also not clear how families utilize public infrastructure in ILS for interactive learning.

In this processing work, we present an initial exploration of the design opportunities for family-based privacy education in ILS. In particular, we focus on the following research questions:

- RQ1: How do families conduct privacy education, and what are the characteristics and challenges involved?
- RQ2: How do family groups utilize ILS for interactive learning?

In the present work, we conducted a preliminary interview study with five families. Each family includes a parent and a child (age 6-14, a critical age range where they can understand privacy concepts [4]). The interviews focused on the roles that each family member played in privacy learning, the approaches, and the challenges, as well as the common practices of utilizing public resources in ILS for interactive learning. Our findings highlighted several characteristics of existing family-based privacy education as well as the challenges. In the meantime, our results also suggested the positive outcomes when utilizing ILS for interactive learning. In the discussion, rooted in our empirical evidence, we re-iterated why family-based privacy education in ILS could be a promising direction to enhance children's learning of privacy, then outlined plans for future work, such as framing design implications, co-design with families to come up with design prototypes, and so on.

## 2 DATA COLLECTION AND ANALYSIS

To inform the design of family-based privacy education interfaces in informal learning settings, we conducted a preliminary semi-structured interview study to understand (1) families' existing practices to address privacy challenges and carry out privacy education, and (2) how families utilize informal learning spaces for family-based learning. We present the details below. This study is approved by our university's Institutional Review Board.

### 2.1 Participants Selection

As our study focuses on family-based privacy education, we aim to look for families that meet the following criteria: (1) the family must have a child between the ages of six and fourteen; (2) the child in the family should have some experience with intelligent technologies and the Internet; (3) the families should have prior experiences visiting the following places: museums, science centers, zoos, aquariums, botanical gardens, nature centers, or historical sites; and (4) each family should have at least one child and one parent present during the interview session.

We chose children between six and fourteen for two reasons: (1) given the complexity of the concept of privacy, children would not start to develop the cognitive ability to comprehend and engage with such complex concepts until the age of six [4, 9, 33];

(2) psychologically, children between the ages of six and fourteen are at a critical time of development to acquire a wide range of knowledge and skills which may further influence their behaviors and decision-making processes [31];

As an ongoing work, we recruited our participants mainly by posting our recruitment materials on social media (e.g., Twitter and Facebook), then used snowball sampling on the selected participants (i.e., we asked the participants to help us identify other families who may be interested in participating in our research). Up to now, we interviewed five families, with one parent and one child in each family group. They all lived in the USA and represented a wide range of backgrounds, ages, and geographic locations. Table 1 in the Appendix summarizes participants' demographic information.

### 2.2 Interview Protocol

To account for the two different participant groups, we prepared two interview protocols, one for the parents and one for the children. The interview questions were consistent in both versions, although certain phrasings were modified to ensure comprehension by the children [27]. The complete interview protocols for both parents and children can be found in the Appendix A.1. The interview protocol contains three sections: experiences and concerns regarding privacy issues with family context, specific needs of family members concerning learning and managing privacy, and their experiences with informal learning spaces

### 2.3 Interview Procedure

All interviews were conducted over Zoom, with parents and children being interviewed separately. When the interviews started, parents were asked to read and sign the consent form while children were asked to read the consent and assent forms together with their parents, then signed them. Then, we obtained participants' oral consent for audio and video recording and then turned on Zoom's live transcription feature for transcription. Next, we proceeded with the interview questions and asked follow-up questions as needed. Each interview was conducted by two co-authors, with one primarily asking the questions while the other focusing on note-taking. Interviews with parents lasted from 30 to 55 minutes (average = 43), and interviews with children ranged from 23 to 59 (average = 42). Upon completion, each participant received a compensation of \$20 (i.e., the family group with one parent and one child would receive \$40).

### 2.4 Data Analysis

We transcribed all audio data and conducted a thematic analysis [3]. Two co-authors read the interview transcripts several times and coded a sample of the data collaboratively to generate two initial codebooks, one for children and one for parents. Using the initial codebooks, the two co-authors code the rest of the data individually. They constantly compared and discussed their codes and resolved any disagreements as they coded, then updated the codebooks as needed. Upon completion of the coding, the two co-authors cross-checked each other's coding again to ensure full agreement. Then, the co-authors examined the codebooks, discussed them, and grouped the codes into higher-level themes.

## 2.5 Interview Research Ethics

As our research involved minors between the ages of six and fourteen, we paid extra attention to our research ethics. First of all, we ensured that all parents and children were well informed of the study procedure, their rights, and measures they could take in the case of having to withdraw through IRB-approved consent and assent forms. Second, similar to McReynolds et al. [22], interviews with parents and children were conducted separately. This was to avoid interference between participants. To ensure the transparency of interviews and reduce the safety concerns of the parents, similar to Sun et al.'s work [32], parents were allowed to observe their children's interviews if they wished to. Finally, we took the following measures to ensure the confidentiality of participants' sensitive personal information: (1) during the interviews, we consistently reminded participants to focus on their own experiences and opinions and not reveal private or sensitive information throughout the interviews; (2) we stored and handled data securely, used pseudonyms or participant codes instead of real names, and anonymized any identifiable information during analysis and reporting; (3) for our children participants, we employ child-friendly language to remind them to keep their privacy and ask questions about their perceptions.

## 3 INITIAL RESULTS

In this section, we present our initial findings. Given the qualitative nature of this research, we intentionally avoided reporting the frequency of all quotes.

### 3.1 Characteristics of Existing Family Privacy Education

**3.1.1 Children Played Multiple Roles in Privacy Education.** In family-based contexts, children's engagement in learning about privacy has been largely expanded, thus playing multiple roles.

**Children as a Vulnerable Group.** As also mentioned in prior work [14], some parents took precautionary measures and set up certain rules for their children, such as not inviting strangers into their homes and refraining from disclosing their SSN and birth date to others. Most parents also admitted that they regularly checked in with their children to ensure they were safe, which included monitoring their children's use of social media (P01, 35-44, M).

**Children as Active Learners.** In our ongoing study, most children actively engaged in learning privacy. Children gained privacy knowledge through family educational activities, covering general topics and privacy-focused education. This education included body/physical privacy, information privacy, and family privacy (e.g., keeping home addresses confidential). Parents used methods like discussions and situational simulations to help children understand and manage privacy (P03, 35-44, M).

**Children as Privacy Experts.** Interestingly, when viewing privacy education at the family level, many children have also played a new role as privacy experts and took the responsibility to teach other family members about privacy. Some children shared the privacy protection methods (particularly regarding online privacy) that they found useful with their family members, such as their parents and siblings. C03 (12-14, M) said, *"I taught my sister to change the password every three months ..... She finds it stressful. She says,*

*oh, nobody's trying to actually interact. So instead, I make her change the passwords every three months, as opposed to monthly, but I do mine. And now she's getting used to it."* (C03)

**3.1.2 Privacy Education Was Often Carried Out through Open Conversations.** Among the families we interviewed, open conversations between parents and their children remained the primary and one key channel for communicating privacy. Oftentimes, parents and children openly discussed their privacy experiences and concerns, collaboratively learned about privacy, and managed children's privacy. P01 (35-44, M) said, *"My boy is frequently on social media. I always have conversations with him, whether it's online or offline. There's a limit to what you can share out there."* (P01)

Parent-child conversations conveyed privacy knowledge and fostered the establishment of family privacy norms. As many parents reported, during early childhood, parents imposed rules for their children to follow. As children grow, parents started to communicate with them and guide them in understanding and managing privacy, such as discussing how to protect privacy and delineating permissible actions. This interactive process involved parents expressing privacy concerns, explaining decisions, and allowing children to articulate their perspectives. Through reaching agreements, both parties gained a comprehensive understanding of family expectations, solidifying them as norms.

**3.1.3 Privacy Education Often Happened in Specific Contexts.** Family-based privacy education often happened in-situ in specific contexts, as family-based privacy education often emphasized practicality and the resolution of specific concerns. In proactive privacy education, children primarily learned how to manage privacy issues. As C01 (12-14, M) said, *"My dad always reminds me to ensure that our messages remain private and that secrets, even family secrets, should not be disclosed easily."* (C01) To do this, parents frequently provided hands-on demonstrations of privacy protection features. For instance, P05 (35-44, F) shared an experience where she discovered her child texting a stranger. To enhance her child's privacy awareness, she simulated an online interaction, posing as a stranger and eliciting various information to demonstrate potential consequences.

**3.1.4 Privacy Education Is a Continuous Process.** Unlike privacy through certain materials, family-based privacy education is a continuous process, which is further tailored to the distinct needs and developmental stages of children. In our data, parents commonly initiated privacy awareness and management with their children when they were young, starting with physical and spatial privacy (P01, 35-44, M; P04, 35-44, M). As children grow, parents often started to impart knowledge about online privacy, covering aspects like social media and passwords. P03 (35-44, M) mentioned, when his child grew older, the timing of privacy education became critical, *"I have to choose the right time. That's very, very important. And I need to have a vague idea of what I'm going to discuss with them."* (P03)

### 3.2 Challenges of Existing Family Privacy Education

**3.2.1 Tensions among Different Family Members.** One prominent challenge among our participants related to the tensions among

different family members, which remained a significant challenge in family-based privacy education.

**Crossing each other's boundary.** During the early stages of childhood, when children lack a comprehensive understanding of privacy, they violated other family members' boundaries. P02 (25-34, Non-Binary) shared an instance where his little daughter frequently invited friends for overnight stays, causing inconvenience to other family members. Parents also violated their children's privacy, especially as they grow older and come to understand privacy and carry secrets they don't want to reveal to their parents.

**Overprotection may cause surveillance concerns.** Some parents seek to maximize child protection, leading to privacy invasions like trying to install cameras in their rooms (P04, 35-44, M) and asking for passwords (P01, 35-44, M). This surveillance often makes children feel their privacy is undervalued, fostering resistance and prompting them to protect it by withholding information and bypassing parental oversight. P01 (35-44, M) shared about his second account, *"My dad has a parental control on my phone. So he sees most of the test. He knows about about a on my smartphone. I have a second account only on my tablet because my dad does not know about my second account."* (P01)

**3.2.2 Complex Contextual Nature of Privacy.** Family privacy education faces a key challenge due to the contextual nature of privacy [26]. The complex contexts of privacy issues in daily life present significant obstacles to conducting effective educational initiatives. As P04 (35-44, M) said, *"I feel (teaching) privacy is slightly more challenging because it's closely tied to his (child's) experiences in life"* (P04). P04 highlighted a key challenge: the impact of advancing technology. Each technological development introduces distinct privacy challenges, demanding an ongoing reassessment of educational approaches. However, many parents lacked the resources to stay informed. Privacy concerns varied in different contexts—home, school, and online spaces.

**3.2.3 Limited Educational Approaches.** As mentioned in Section 3.1.2, family-based privacy education often focused on open conversations. Yet, this approach faces challenges, particularly in parent-child dialogues leading to emotional conflicts. P02 (25-34, Non-Binary) noted children's occasional strong resistance, hindering effective communication and causing frustration. They stated, *"Issue you experience in education about privacy is, just being rebellious, initially."* (P02) Most educational discussions mainly revolved around privacy rules and potential negative outcomes. Children, expressing boredom and displeasure in interviews, found this repetitive method unappealing. Therefore, diverse approaches are essential for privacy education within families.

**3.2.4 Lack of In-Depth Knowledge among All Family Members.** Parents and children lack sufficient educational support for enhancing privacy literacy. Parents educate children about privacy based on their backgrounds, experiences, discussions with other parents, and limited online research. Some parents admit inadequate knowledge, hindering in-depth conversations and comprehensive privacy guidance. Consequently, parents resort to social media and online resources for additional information. P03 (35-44, M) shared, *"Most of the time, I meet challenges in educating my children about privacy. I need to have a vague idea of what I'm going to discuss with them."*

*That's why I do my research on various topics such as identity theft, social media, cyberbullying, and everything related to that. So I can have an in-depth understanding of what I want to discuss with them. In case they have questions, I will be able to answer their questions and put their concerns to rest."* (P03) Unfortunately, existing resources often lack organized privacy education to improve parental privacy literacy. These materials often lack precision, covering general privacy rules instead of specific details relevant to privacy literacy.

### 3.3 Family-based Learning Experiences in ILS

One of the main motivations of our study is to understand whether and how families utilized ILS for family-based learning, which may potentially provide new possibilities for family-based privacy education. The findings from our interviews indicated a widespread acknowledgment among both parents and children regarding the efficacy of ILS.

**3.3.1 ILS Warranty Quality Family Time.** Families visiting various ILS tend to enjoy quality time together. During the visit, parents and children explore together, taking photos (P02, 25-34, Non-Binary), interacting with exhibits in museums or science centers (e.g., P03, 35-44, M; P04, 35-44, M; P05, 35-44, F), and actively participating in discussions.

In these settings, family conversations played a crucial role in interaction. Both parents and children could start conversations. Typically, parents began by asking their children for opinions on exhibits, models, or books, setting the stage for deeper exploration. They also enhance their children's experience by researching the places beforehand (e.g., P04, 35-44, M). As C04 (9-11, M) said, *"Sometimes my parents will go like, what did you learn? Like, something like that?"* (C04) C05 (12-14, F) shared, *"My mom will sometimes like, tell me oh, the motion of the tide or something? Or how seeds get transported from place to place? But yeah."* (C05)

From the children's perspective, they often shared novel experiences with their parents, seeking to gain more knowledge about these matters through ensuing discussions. P04 (35-44, M) provided an example in which his child was able to ask many questions related to the exhibits, *"He would say, 'Dad, why is the electromagnetic situation like this? Why does static electricity make my hair stand up?' We would then discuss how static electricity transforms into attraction, and so on."* (P04) At times, similar to the phenomenon we observed in Section 3.1.1, children also took on the role of teachers and proactively engaged in discussions with their parents. P04 (35-44, M) continued to give us an example in which his child became so obsessed with the exhibit in a local science center, *"My child said 'Dad, I understand. Should I teach you this?'"* (P04)

**3.3.2 Enriching Learning.** Parents and children in our study saw ILS as a platform for interactive and creative learning. The various interactive elements, like designs and experiments, offered different ways to grasp concepts and motivated children to explore independently, sparking curiosity. Kids particularly enjoyed engaging in activities like painting and crafting at ILS, which not only brought joy but also prompted them to think beyond the activities. For instance, C05 (12-14, F) shared, *"There also like recreations of models, such as train cars, etc. That sort of made me think about it a little more and view it in a way I hadn't before about the Holocaust. So I think"*

*that applies to every sort of museum.”* (C05) Moreover, ILS organized many events for visiting children, offering chances to earn awards and fostering a sense of accomplishment, thus encouraging ongoing exploration. For instance, C02 (12-14, M) shared her experience, *“The most satisfying experience for me in these places was winning an award for the best night grader who ever used the museum. When you get rewarded for something you’ve done, it motivates you to strive for more and achieve greater things.”* (C02)

**3.3.3 Ongoing Influence in Life.** Our study found that ILS significantly impact long-term family learning experiences beyond individual visits. This enduring influence arises from the flexible and dynamic nature of informal learning, seamlessly integrated into everyday life. For example, families, post-ILS visits, engage in ongoing discussions, especially prompted by activities experienced during their visit. P03 (35-44, M) described, *“After the visit, we usually discuss things like ‘What did you like most about this visit?’ And it was during this time that we explored different cultures and went to see the dome... So we all had a thorough discussion about it ...”* (P03) Such ongoing influence was embedded in everyday life and often occurred seamlessly and naturally.

## 4 DISCUSSION AND FUTURE WORK

We proposed the concept of *family-based privacy education in information learning spaces* as a promising approach to enhancing children’s learning of privacy and improving their privacy literacy. This is different from the mainstream children-centered privacy education in the literature. In this section, we first reflected on the insights we gained in our preliminary interview and discuss why the proposed approach has the potential to transform the landscape of privacy education for children. We also outline future work, which involves framing design opportunities for interfaces supporting family-based privacy education in ILS and co-designing design prototypes with family groups.

### 4.1 Why Family-based Privacy Education May Work in ILS?

Our data suggested that family-based privacy education within ILS may serve as a bridge to fill gaps present in the current children-centered privacy education. Informal learning spaces possess three key characteristics: warranty quality family time, enriching learning experiences, and ongoing influence. These aspects not only accentuate the existing strengths of family privacy education but also address its limitations.

For example, quality family time in ILS provides an opportunity for children to adequately and actively play their role in the learning process as well as engage in open conversation about privacy with their siblings and parents, which may further help mitigate the tension among different family members. Additionally, such interaction may extend beyond families’ visits to ILS and continue to spark relevant conversations within the family. The long-term engagement further guaranteed that children would have the opportunity to continue discussing privacy and related topics with their parents in different contexts, which would be closer to the contextual nature of privacy [26]. Similarly, when privacy education happens in a specific context in ILS (e.g., an interactive exhibit

that visualizes how data flows among different devices and entities in a smart home setting), it would spark several “teachable moments”. For instance, in the example of an interactive exhibit that visualizes data flows within a smart home, when the children discover that some information is being sent to a mysterious third-party server, it is a “teachable moment” to educate them about the concept of “third-party tracking”, “ad network”, and other concepts. These teachable moments would provide specific scenarios for children to contextualize their learning, help them to understand the complex nature of privacy more easily, and at the same time, provide deeper knowledge that reveals the facts behind the observable phenomenon on the surface.

### 4.2 Future Work

Faced with increasing privacy risks for children and constraints of existing privacy education, we introduced a **family-based privacy education approach in informal learning spaces**. An interview study involving five families confirmed the viability of this approach in understanding the characteristics and challenges of current family privacy education, as well as how families utilize informal learning spaces for learning.

As a work in progress, we will enhance diversity by incorporating more family groups with different backgrounds. For children, we also will adjust interview protocols for varied ages to help them engage in interviews. Then we also consider collaborating with more stakeholders, such as educators and personnel from diverse informal learning spaces. This broader inclusion will provide richer insights into families’ varied experiences and needs in different contexts. We will build upon existing work, such as family-based learning initiatives focusing on AI, digital literacy, and privacy education [10, 19, 36, 39]. Additionally, enhancing connections with learning sciences in related literature and providing detailed data analysis will enhance future work. Based on this understanding, we will also actively engage with family groups in co-design activities to obtain design implements for privacy-related educational devices within informal learning spaces. Furthermore, we intend to design and implement interventions based on our findings, with a commitment to continually refining and improving the effectiveness of these interventions. This approach ensures a dynamic and responsive strategy in addressing the complex challenges of privacy education within family settings and informal learning spaces.

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## A APPENDIX

### A.1 Interview Protocols

#### A.1.1 Interview Protocols with Parents. 1 Warm up

- How would you define privacy within the context of your family and daily life?
- Do you ever think of privacy offline?
- Do you ever think of privacy online?
- Do you think there are any privacy challenges in your life?
- Can you share a little about your family's daily routines and activities?

#### 2 Experiences and concerns about family-based privacy problems

- Have you faced any privacy-related situations or dilemmas within your family?
- How do you and your family feel about privacy within your household?
- Are there any specific concerns or challenges you face, especially for children?
- Have there been any conflicts among your family members regarding privacy?

#### 3 Existing educational methods / practice related to these family-based privacy problems

- What measures do you or your family usually take to address privacy issues? Can you share an impressive experience with us?
- Do you believe these measures are sufficient to handle privacy issues?
- Do your family members discuss private matters with each other?
- Are there any specific topics or aspects of privacy that you focus on when discussing it with your children?
- Do you or your family help other families manage privacy issues in daily lives?

#### 4 Needs of family members in privacy learning and managing in informal learning spaces

- How do you handle situations where your child's privacy is violated or compromised?
- Have you encountered any challenges or difficulties in educating your children about privacy or talking about privacy with your children?
- What are your hopes or expectations about privacy within your family?
- How do you and your family usually learn about technology-related topics, such as artificial intelligence, privacy, and information technology?

#### 5 Experience in informal learning spaces

- Have you studied or learned together with your children outside of school?
- Among all the places you have visited, which one is your favorite one?
- What do you usually do with your children there?
- How do you feel about learning privacy-related topics with your children in these places, such as museums, science centers, or libraries and so on?

- How do you think these experiences in those spaces have influenced your family's overall interest in learning and curiosity?
- In your opinion, do these spaces, just like museums, science centers and libraries, can be a good place for families to learn about privacy?

#### A.1.2 Interview Protocols with Children. 1 Warm up

- Can you tell us a bit about yourself and your family?
- Do you have any secrets?

#### 2 Experiences and concerns about family-based privacy problems

- What are your secrets about? You don't have to tell me the exact information because it's your secret, but can you give me an idea of what it might be?
- Have you ever felt like someone is watching your secret too much or that things you don't want others to know are too easy for them to find out?

#### 3 Existing educational methods/practice related to these family-based privacy problems

- How does your family keep your secrets, like your own special box?
- How do you protect your secret?
- Have you ever told someone in your family when you felt like your secret was not safe?

#### 4 Needs of family members in privacy learning and managing

- Do you think you need more help to keep your secret safe and make sure others don't know too much about it?
- What kinds of technology do you use in your everyday life?
- How did you learn to use these things like smartphones or tablets?
- Do you ever talk with your friends or family members about the things you learn or do with technology?

#### 5 Experience in informal learning spaces

- Have you ever visited these places like museums or science centers to learn about new things?
- Can you tell me about the most fun thing you've done at a museum or science center?
- What was your favorite exhibit or activity at the museum or science center?
- Did you talk about what you learned at the museum or science center with your family or friends?
- If you were a designer, how would you design these places for children?

### A.2 Participants' demographics

| ID  | Age   | Gender     | Education  | Employment Status | Informal Learning Spaces                                              | Frequency    | Specific Names                                                          |
|-----|-------|------------|------------|-------------------|-----------------------------------------------------------------------|--------------|-------------------------------------------------------------------------|
| P01 | 35-44 | Male       | Bachelor's | Full-time         | Museums, Science Centers, Historical Sites, Libraries                 | Occasionally | Local Museum of Arts, Local Science Center, Local Library               |
| C01 | 12-14 | Male       | 7th-9th    | N/A               |                                                                       |              |                                                                         |
| P02 | 25-34 | Non-binary | Bachelor's | Full-time         | Museums, Science Centers, Libraries                                   | Occasionally | Local Science Gallery, Local Library, National Museum, National Library |
| C02 | 12-14 | Male       | 7th-9th    | N/A               |                                                                       |              |                                                                         |
| P03 | 35-44 | Male       | Bachelor's | Full-time         | Museums, Science Centers, Historical Sites, Libraries                 | Occasionally | Local Cultural Center, Local Museum of Science and Industry             |
| C03 | 12-14 | Male       | 7th-9th    | N/A               |                                                                       |              |                                                                         |
| P04 | 35-44 | Male       | Graduate   | Full-time         | Museums, Science Centers, Nature Centers, Libraries                   | Occasionally | Local Science Center                                                    |
| C04 | 9-11  | Male       | 4th-6th    | N/A               |                                                                       |              |                                                                         |
| P05 | 35-44 | Female     | Bachelor's | Homemaker         | Museums, Science Centers, Nature Centers, Historical Sites, Libraries | Occasionally | Discovery Cube, Missions, Local Natural Museum, Local Science Center    |
| C05 | 12-14 | Female     | 7th-9th    | N/A               |                                                                       |              |                                                                         |

**Table 1: Participants demographics.**